

Solution and 6-week delivery plan

A secure, cloud-based patient records system for small clinics in India, from front desk to prescription.



6 weeks

From kick-off to pilot go-live

13 modules

From front desk to data export

5 clinics

Live at the end of week 6

Based on

Statement of Work v1.0 and the Phase 0 technical assessment

Version

2.1, 15 September 2026

Status

For client review and approval

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Section 1

Summary

We will design, build, secure and launch the complete clinic EMR in 6 weeks, with 5 pilot clinics live at the end of week 6.

Timeline

6 weeks

Kick-off to pilot go-live

Pilot

5 clinics

Live in week 6

Running cost (pilot)

₹23,500 / mo

Cloud, monitoring, messaging

Needed before day 1

4 decisions

See Section 11

What you get

Front desk

- Clinic setup
- Patient registration
- Appointments & live queue
- Billing (invoice + payment)

Clinical

- Patient summary screen
- Consultation notes
- Prescriptions with PDF
- Reports & documents

Communication & data

- WhatsApp inbox & reminders
- Secure share links
- Reports dashboard
- Data import & full export

How 6 weeks is achievable

We deliver in 6 weeks by running five build streams in parallel from day 1, with a checkpoint at the end of every week. This depends on the client providing the four key decisions before day 1 and giving approvals within 48 hours.

Built-in safeguards

Safeguard	What it means for the clinic
Clinic data is fully separated	One clinic can never see another clinic's patients, enforced by the database itself
Private notes stay private	Doctors' internal notes can never be sent to a patient
Records are never overwritten	Finalised records are locked; corrections are added as new entries
Every action is recorded	A permanent log shows who viewed or changed what, and when
Data stays in India	All data is stored and processed in AWS Mumbai
No lock-in	Clinics can download all their data at any time, without asking us

Section 2

Scope

All modules in the Statement of Work are included. The table shows what each one covers at pilot launch.

Module	Included at launch	Notes
Clinic setup	Clinic profile, locations, services and fees, users and roles	—
Patient registry	Registration, search by mobile or name, duplicate check, merge	Search before create is enforced
Appointments & queue	Booking, walk-ins, live queue, reordering, statuses	—
Patient summary	Allergies, conditions, current medicines, recent visits	Allergies shown first
Consultation	Notes, vitals, diagnoses, templates, copy from last visit	Autosaves on weak internet
Prescriptions	Drug search, dosing shorthand, allergy warnings, signed PDF	Pilot drug list, see Section 7
Reports & documents	Upload, virus scan, view, share	—
WhatsApp	Inbox, prescription delivery, reminders, opt-out	Depends on Meta approval, see Section 6
Automation	Appointment, follow-up and missed-visit reminders	—
Billing	Invoices and payments	No accounting or GST filing
Reports dashboard	Visits, collections, appointments	—
Data import	Patient list from CSV / Excel	Legacy system migration quoted separately
Data export	Full clinic data with original files	Self-service

Not included

Inpatient / ward management, pharmacy stock, full accounting and GST filing, lab device integration, insurance claims, native mobile apps, video consultation, ABDM certification. The system is designed so these can be added later.

Section 3

Solution overview

A web application that works on any modern browser, hosted entirely in AWS Mumbai.

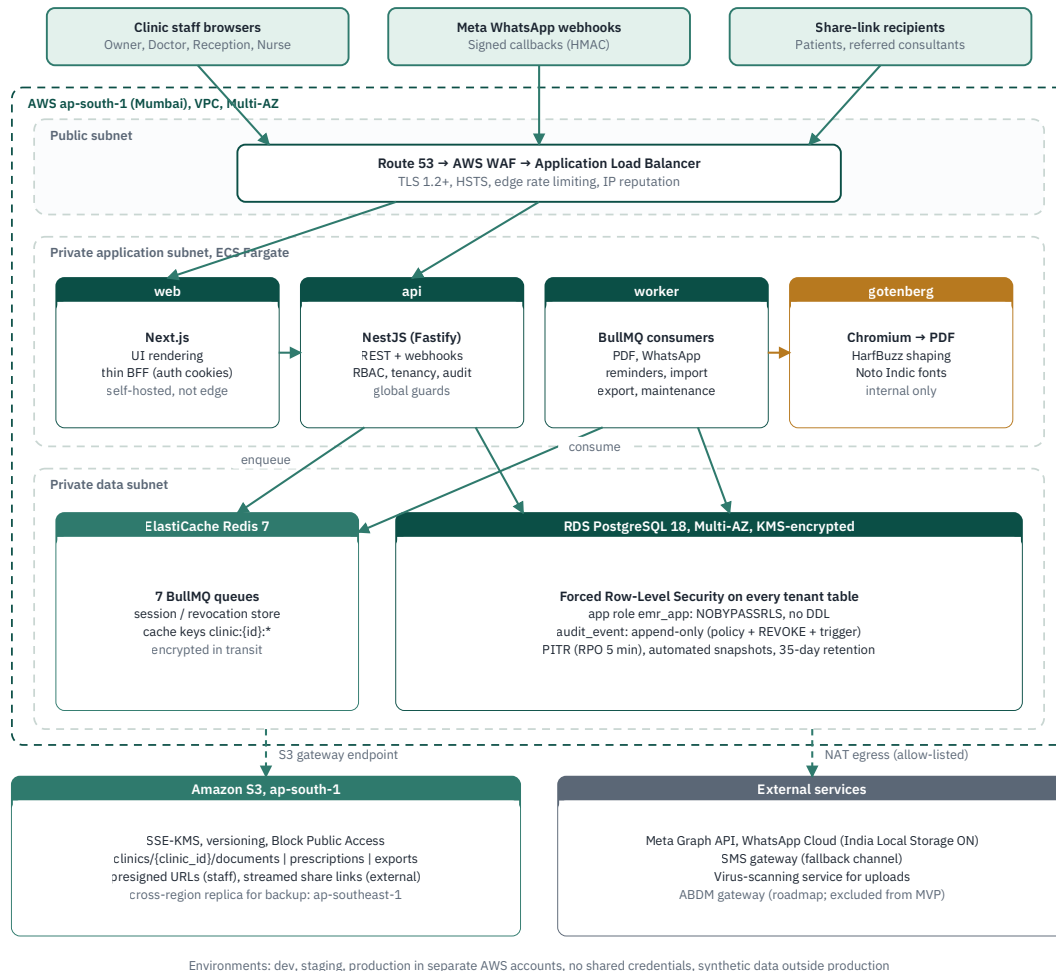


Figure 3.1, System architecture. Only the secure entry point is public. Application, database and files sit in private networks.

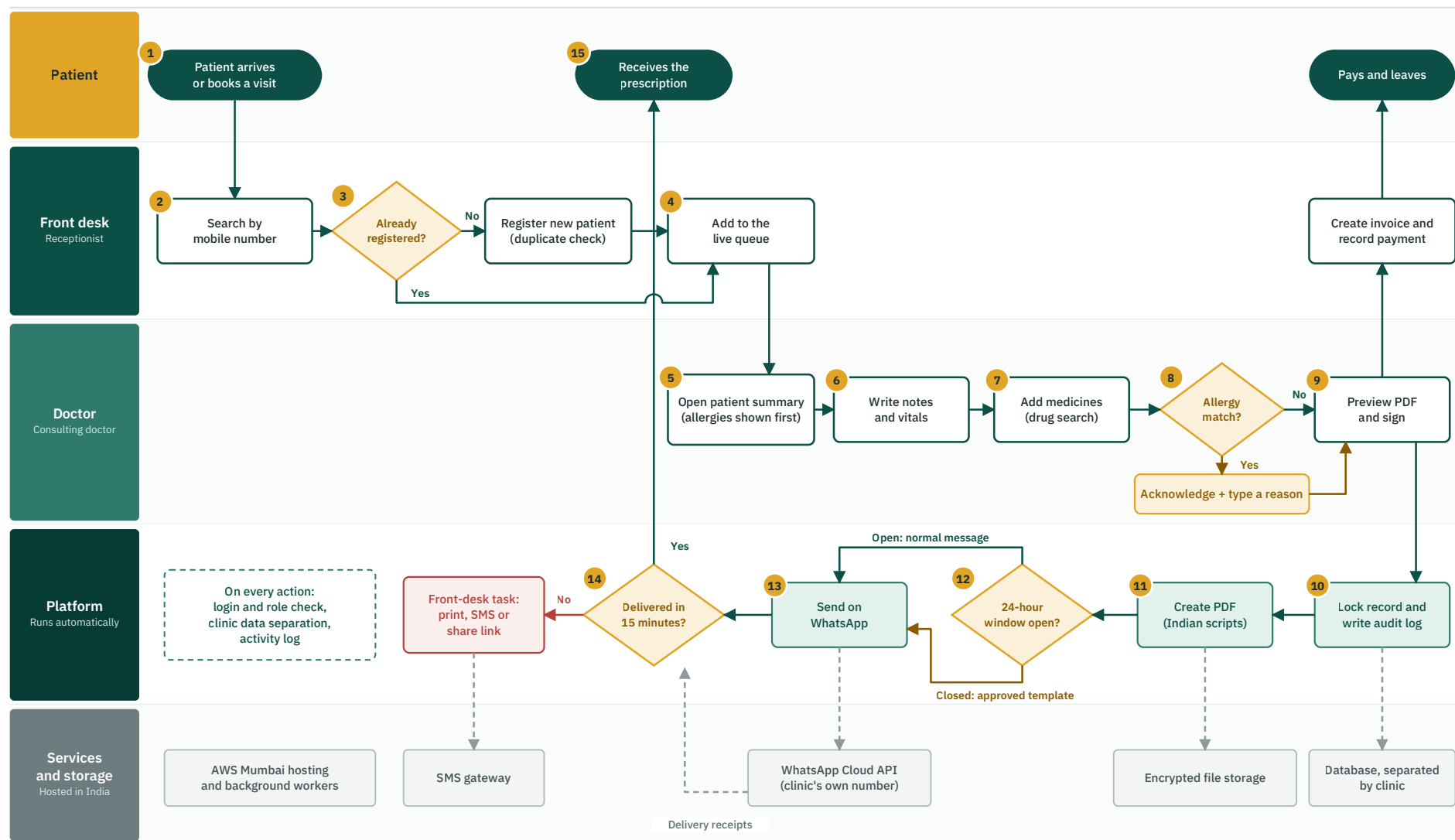
Technology

Layer	Technology	Why
Screens	Next.js, React	Fast, responsive web app for desktop and tablet
Server	NestJS (Node.js, TypeScript)	Security checks applied in one place for every request
Database	PostgreSQL	Reliable, with built-in clinic-level data separation
Files	Amazon S3 (Mumbai)	Encrypted, private document storage
PDFs	Chromium-based renderer	Correct Hindi, Tamil and other Indian scripts
Messaging	Official WhatsApp API	Each clinic uses its own WhatsApp number
Hosting	AWS Mumbai	Data stays in India

Section 4

How the system works

One patient visit, end to end: who does what, what the system does automatically, and where each step is stored. Numbers follow the main path.



Start or end of the visit Step done by staff Step done by the system Decision point Hosted service 1 Order of the main path

Section 5

Security and data protection

Designed for India's DPDP Act 2023 and DPDP Rules 2025, with security enforced by the system rather than by staff discipline.

Area	How it is handled
Login	Strong password hashing; two-factor authentication for owners and doctors
Sessions	Short sessions; access removed instantly when a staff member leaves
Permissions	Five roles; anything not explicitly allowed is blocked
Encryption	All data encrypted in transit and at rest
Audit trail	Permanent, tamper-proof log of every sensitive action
Consent	Separate consent for treatment, data use, WhatsApp, marketing, sharing and ABDM
Backups	Continuous backup; at most 5 minutes of data loss; restored within 4 hours
Testing	Automated checks on every change that one clinic can never reach another's data; external security test before launch

Who can do what

☒ Full access
 ☐ View only
 ☐ No access

Area	Owner / Admin	Doctor	Receptionist	Nurse	Auditor
Clinic settings & users	Full	View	View	View	View
Patient details	Full	Full	Full	View	—
Appointments & queue	Full	Full	Full	Full	—
Consultation notes	View	Full	—	View	—
Finalise & sign prescription	—	Full	—	—	—
Private doctor notes	—	Full	—	—	—
Vitals & allergies	View	Full	—	Full	—
Documents & sharing	Full	Full	Full	View	—
WhatsApp inbox	Full	Full	Full	Full	—
Billing	Full	View	Full	—	—
Reports	Full	View	—	—	View
Audit log	View	—	—	—	View
Import & export	Full	—	—	—	—

Only doctors with a registration number on file can sign prescriptions. Receptionists see visit times but not clinical notes. Auditors see the activity log, not patient records. Full data export is limited to the Owner/Admin.

Section 6

WhatsApp communication

Each clinic connects its own WhatsApp Business number, so patients receive messages from their clinic.

What it does

- Sends prescriptions and reports as secure links
- Appointment and follow-up reminders
- Shared inbox for patient replies
- Honours STOP / opt-out immediately

If a message fails

- Front desk gets a task within 15 minutes
- Print, SMS or share link as fallback
- Clinical records are never affected

What we need

- Meta business verification (1–3 weeks)
- One WhatsApp number per clinic
- Approval of message templates

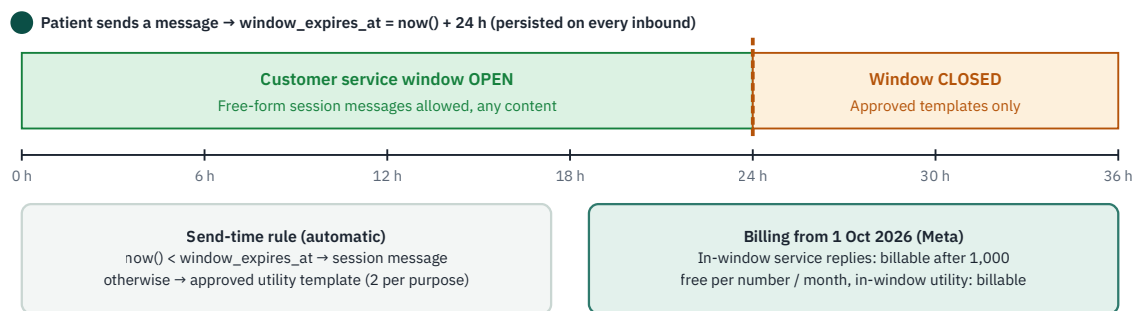


Figure 6.1, WhatsApp 24-hour rule. Free-form replies are allowed for 24 hours after a patient's message; after that, only pre-approved templates can be sent. The system switches automatically.

Estimated messaging cost

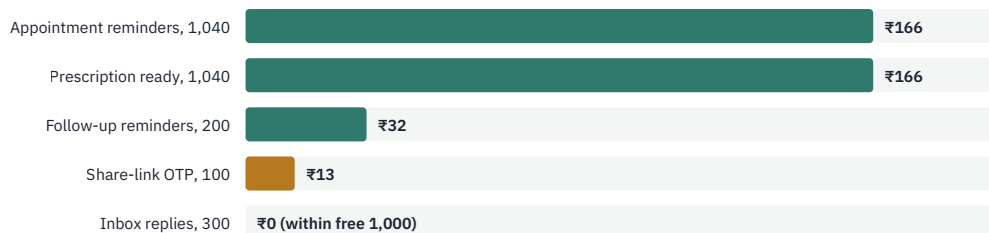


Figure 6.2, About ₹377 per clinic per month for a clinic seeing 40 patients a day. This is paid to Meta. Rates are indicative and should be confirmed against Meta's current India pricing.

Recommendation

Include 2,000 messages in the clinic subscription and charge about ₹0.30 per message beyond that. Usage is visible to each clinic in the admin screen.

Section 7

Prescriptions and drug data

A full drug-interaction database requires a paid licence. The pilot launches with a curated drug list and allergy checking.

Option	Cost	Includes	Status
Pilot drug list (~2,000 medicines)	Included	Drug search, allergy warnings, duplicate-medicine check	At launch
Licensed Indian drug database	₹1–4 L per year	Adds drug interactions, contraindications, child dosing	After pilot

Safety features at launch

Included

- Allergy warning that must be acknowledged, with a typed reason to override
- Duplicate medicine warning
- Schedule X restriction for teleconsultation
- Free-text entry for any medicine not in the list
- Preview of the exact PDF before signing

Added with the licensed database

- Drug-to-drug interactions
- Condition-based warnings
- Weight-based child dosing
- Pregnancy warnings

Important

Interaction checking will **not appear on screen** until the licensed database is in place, so doctors never assume a check is running when it is not. Switching to the licensed database takes about 1–2 weeks.

Doctor-friendly prescribing

Doctor types	Patient sees on the prescription
1-0-1	1 tablet in the morning and 1 at night
0-0-1	1 tablet at night
SOS	Take only when needed

Prescriptions carry the clinic letterhead, doctor name, qualifications, registration number and signature block. Aadhaar eSign can be added after the pilot.

Section 8

Data import and export

Clinics can bring their patient list in and take all their data out, at any time.

Import



No row is dropped silently. Any row with a problem is listed with the reason, so the clinic can correct it and upload again.

Export

Included in the export package
Patients, appointments, visits, vitals, diagnoses, allergies, prescriptions
All uploaded documents (original files), communication records, billing, activity log
A file index with checksums and a plain-language guide to every file

The owner starts an export from the admin screen. The download link is valid for 7 days, and every export is logged.

Section 9

6-week delivery plan

Five build streams run in parallel from day 1, with a clear checkpoint at the end of every week.

Workstream	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Design	Design system, clickable prototype	Test with 5 clinics	Screen design support			Training material
Platform & security	Cloud setup, login, roles, database	Data separation, audit log	Monitoring, backups	Performance tuning	Security test, restore test	Production launch
Front desk	Clinic setup	Patient registry	Appointments, queue	Billing	Reports dashboard	Fixes
Clinical	Consultation screen	Patient summary	Prescriptions, PDF	Documents, corrections	Doctor feedback fixes	Fixes
WhatsApp	Meta verification starts	Inbox, templates submitted	Reminders	Share links, opt-out	Failure handling	Live on clinic numbers
Data	Offline autosave	Import wizard	Duplicate checks	Export package	Fixes	Import pilot data
Testing	Test plan, test data	Automated tests every week			Full regression	Launch sign-off
Milestone	Kick-off	Prototype approved	Clinical flow complete	Feature complete	Code freeze	Pilot go-live

Weekly checkpoints

Week	Checkpoint	Client sees
1	Kick-off complete	Clickable prototype, cloud environment ready, Meta verification submitted
2	Prototype approved	Results of testing with 5 pilot clinics, registration and queue working
3	Clinical flow complete	Consultation to signed prescription PDF, end to end
4	Feature complete	All modules working in the test environment
5	Code freeze	Security test report, backup restore proven, performance results
6	Pilot go-live	5 clinics live, staff trained, data imported

Prototype approval (end of week 2)

Five doctors and their receptionists try the prototype. Building continues in parallel, and any screen changes are made in week 3.

Must pass	Target
Register a new patient	Under 60 seconds
Spot an existing family member on the same number	5 of 5 users
Notice a patient's allergy before prescribing	5 of 5 users
Complete a consultation and prescription	Under 4 minutes
Usability score (SUS)	68 or higher

After week 6

Period	Activity
Weeks 7–8 (optional)	Hypercare: close support for pilot clinics after launch
After pilot	Licensed drug database, Aadhaar eSign, ABDM certification, onboarding more clinics

Section 10

Running cost

Monthly cost to operate the platform once it is live.

Item	Pilot (5 clinics)	~100 clinics
Cloud hosting (AWS Mumbai)	₹18,000	₹65,000
Monitoring and domains	₹3,000	₹8,000
WhatsApp (paid to Meta)	₹2,000	₹40,000
SMS fallback	₹500	₹8,000
Total per month	₹23,500	₹1,21,000



Figure 10.1, Monthly running cost. At about 100 clinics this is roughly ₹1,210 per clinic per month.

Cost per clinic at scale

Per clinic, per month (~100 clinics)	Amount
Hosting and monitoring	≈ ₹730
Messaging (WhatsApp and SMS)	≈ ₹480
Total	≈ ₹1,210

Not included in running cost

Licensed drug database (₹1–4 L per year, if taken), Aadhaar eSign, ABDM certification and legacy data migration are priced separately when needed.

Section 11

What we need from you

The 6-week timeline depends on these inputs arriving on time.

Decisions needed before day 1

#	Decision	Our recommendation
1	Drug data: pilot drug list now, or licensed database?	Pilot list now; licence after the pilot
2	Prescription signature: signature block, or Aadhaar eSign?	Signature block now; eSign later
3	Scope sign-off: the modules listed in Section 2	Confirm as listed
4	WhatsApp pricing: bundled messages, or clinic's own provider?	2,000 messages bundled + overage

Inputs and deadlines

Input	Needed by
Meta business verification documents	Day 1
Clinic branding and sample prescription formats	Day 1
5 pilot clinics confirmed (doctors + receptionists)	Day 1
A clinical reviewer available 2–4 hours per week	Day 1
Approvals and feedback within 48 hours	Throughout
Approved consent, privacy and patient notice text	End of week 2
Sample patient files for import (anonymised)	End of week 2
Data retention policy from your legal advisor	End of week 4

Other items to confirm

Item	Our proposal
Backup targets	Max 5 minutes of data loss; back online within 4 hours
Unmatched WhatsApp chats	Deleted after 30 days if not linked to a patient
Clinic leaving the platform	Export → 30-day read-only → verified deletion → certificate
Allergy warning	Must be acknowledged, with a typed reason to override
Accessibility	WCAG 2.2 AA for main screens
DPDP compliance	Your legal advisor to complete the impact assessment and records by May 2027

Section 12

Key risks

The main risks of a 6-week delivery, and how we handle them.

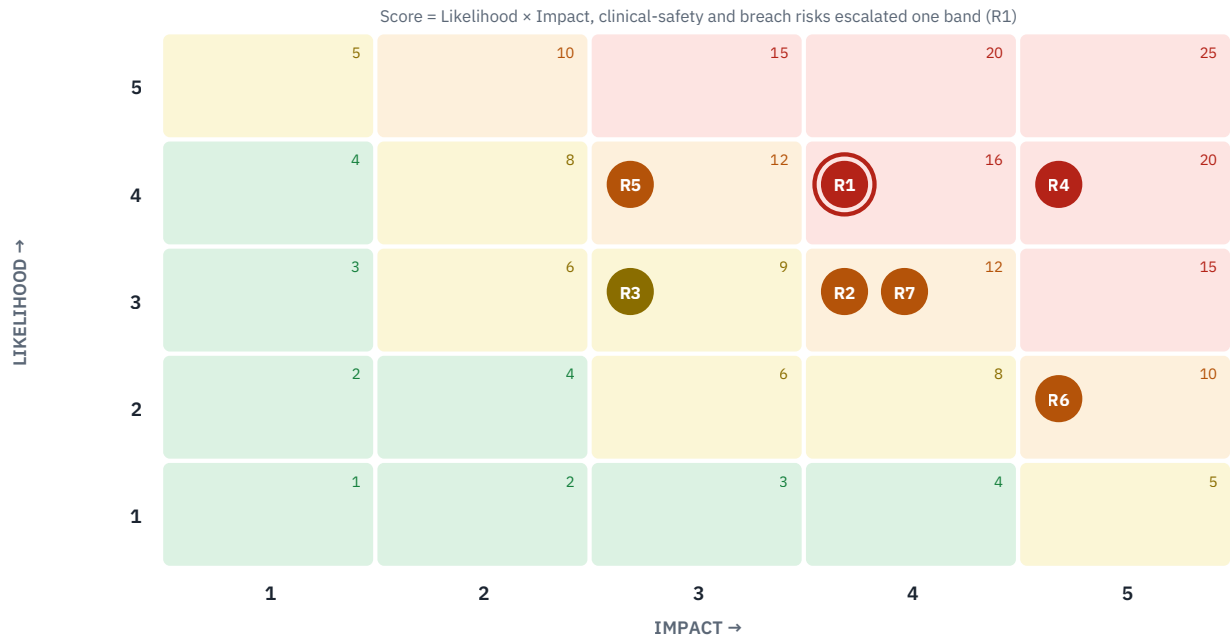


Figure 12.1, Risk map. Higher and further right means more serious.

ID	Risk	How we handle it
R1	Late client decisions or inputs delay the plan	Decisions before day 1; 48-hour approval turnaround; weekly checkpoint
R2	Meta verification takes longer than expected	Submit on day 1; test number meanwhile; print and SMS fallback at launch
R3	Short timeline affects quality	Automated tests every week; code freeze in week 5; external security test
R4	Doctors find the system slower than paper	Prototype tested with 5 clinics in week 2; fixes in week 3
R5	Duplicate patient records	Search before registering; family members shown; manual merge tool
R6	One clinic seeing another's data	Separation enforced by the database; automated checks on every change
R7	Prescription PDF layout or Indian script errors	Tested with long prescriptions and regional scripts; preview before signing

Section 13

Approval

Work starts on day 1 once this plan is approved and the four decisions in Section 11 are confirmed.

Approval	Name / signature	Date
Client / Product Owner		
Delivery Partner		
Scope changes agreed (if any)		

Glossary

Term	Meaning
ABDM	Ayushman Bharat Digital Mission, India's national digital health programme
DPDP	Digital Personal Data Protection Act 2023 and Rules 2025
EMR	Electronic medical record
Hypercare	Close support in the first weeks after launch
SUS	System Usability Scale, a standard usability score
WCAG	Web accessibility standard